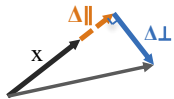


Geometric Decomposition



$$\Delta = \Delta_{\parallel} + \Delta_{\perp}$$

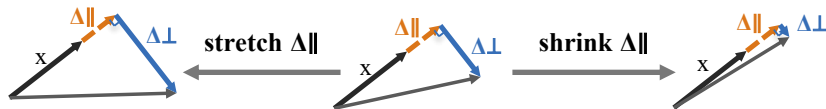
$$\Delta_{\parallel} = \alpha x$$

$$\Delta_{\perp} \perp x$$

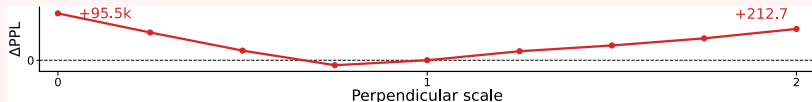
Scaling-like
 $\Delta_{\parallel}$ change magnitude

Direction-changing
 $\Delta_{\perp}$ change direction

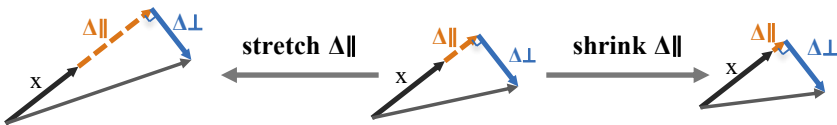
Manipulate the Perpendicular Component



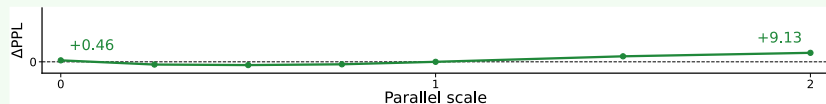
Changing $\Delta_{\perp}$ sharply hurts performance



Manipulate the Parallel Component

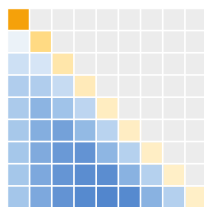


Changing $\Delta_{\parallel}$ has limited impact on performance



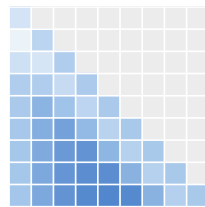
Attention-side View of Parallel Editing

Baseline Attention



Diagonal-suppressed

Diagonal Editing



Parallel edits redistribute diagonal mass